

LAMPIRAN

Lampiran 1 : Aturan *Fuzzy Logic*

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Rendah (Basah)	Dingin	Rendah	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Dingin	Rendah	Redup	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Dingin	Rendah	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Dingin	Sedang	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Rendah (Basah)	Dingin	Sedang	Redup	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Rendah (Basah)	Dingin	Sedang	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Dingin	Tinggi	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Rendah (Basah)	Dingin	Tinggi	Redup	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Rendah (Basah)	Dingin	Tinggi	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Sedang	Rendah	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Sedang	Rendah	Redup	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Sedang	Rendah	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Sedang	Sedang	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Rendah (Basah)	Sedang	Sedang	Redup	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Rendah (Basah)	Sedang	Sedang	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Sedang	Tinggi	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Rendah (Basah)	Sedang	Tinggi	Redup	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Rendah (Basah)	Sedang	Tinggi	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Rendah (Basah)	Panas	Rendah	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Rendah (Basah)	Panas	Rendah	Redup	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Rendah (Basah)	Panas	Rendah	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Rendah (Basah)	Panas	Sedang	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Rendah (Basah)	Panas	Sedang	Redup	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Rendah (Basah)	Panas	Sedang	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Rendah (Basah)	Panas	Tinggi	Rendah (Terang)	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Rendah (Basah)	Panas	Tinggi	Redup	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Rendah (Basah)	Panas	Tinggi	Tinggi (Gelap)	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Sedang (Lembab)	Dingin	Rendah	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Dingin	Rendah	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Dingin	Rendah	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Dingin	Sedang	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Dingin	Sedang	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Dingin	Sedang	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Dingin	Tinggi	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Dingin	Tinggi	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Dingin	Tinggi	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Sedang	Rendah	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Sedang (Lembab)	Sedang	Rendah	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Sedang	Rendah	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Sedang	Sedang	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Sedang	Sedang	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Sedang	Sedang	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Sedang	Tinggi	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Sedang	Tinggi	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Sedang	Tinggi	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Sedang (Lembab)	Panas	Rendah	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Sedang (Lembab)	Panas	Rendah	Redup	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Sedang (Lembab)	Panas	Rendah	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Sedang (Lembab)	Panas	Sedang	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Sedang (Lembab)	Panas	Sedang	Redup	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Panas	Sedang	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Sedang (Lembab)	Panas	Tinggi	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Panas	Tinggi	Redup	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Sedang (Lembab)	Panas	Tinggi	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Tinggi (Kering)	Dingin	Rendah	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Dingin	Rendah	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Dingin	Rendah	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Dingin	Sedang	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Dingin	Sedang	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Dingin	Sedang	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Dingin	Tinggi	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Tinggi (Kering)	Dingin	Tinggi	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Dingin	Tinggi	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Sedang	Rendah	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Sedang	Rendah	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Sedang	Rendah	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Sedang	Sedang	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Sedang	Sedang	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Sedang	Sedang	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Sedang	Tinggi	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Sedang	Tinggi	Redup	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Sedang	Tinggi	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
Tinggi (Kering)	Panas	Rendah	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>

Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Aktuator
Tinggi (Kering)	Panas	Rendah	Redup	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Tinggi (Kering)	Panas	Rendah	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Tinggi (Kering)	Panas	Sedang	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Panas	Sedang	Redup	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Panas	Sedang	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
Tinggi (Kering)	Panas	Tinggi	Rendah (Terang)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Panas	Tinggi	Redup	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
Tinggi (Kering)	Panas	Tinggi	Tinggi (Gelap)	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>

Lampiran 2 : Hasil Pengujian *Fuzzy Logic*

Parameter				Nilai Fuzzy			Aktuator
Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Penyiram	Kipas	Lampu	
1119	18	33	912	0	0	0.85	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1040	19	35	2755	0	0	0,75	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1094	17	31	3838	0	0	0.95	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1119	15	55,4	912	0	0	0	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1040	18	56	2755	0	0	0,1	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1094	19	57	3838	0	0	0.83	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1119	17	84,3	912	0	0	0	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1040	15	80	2755	0	0	0,1	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1094	18	79.9	3838	0	0	0.83	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1119	28,2	33	912	0	0,32	0.85	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1040	28,2	35	2755	0	0,32	0.75	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1094	28,1	31	3838	0	0,31	0.95	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1119	28,2	55,4	912	0	0,31	0	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1040	28,2	56	2755	0	0,32	0.1	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1094	28,1	57	3838	0	0,31	0	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>

Parameter				Nilai Fuzzy			Aktuator
Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Penyiram	Kipas	Lampu	
1119	28,2	80,3	912	0	0,32	0	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1040	28,2	80	2755	0	0,32	0,104	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1094	28,1	79.9	3838	0	0,31	0.83	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1119	31.3	33	912	0	0,63	0.85	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1040	30,2	35	2755	0	0,52	0.75	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1094	32	31	3838	0	0,7	0,95	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1119	31.3	55,4	434	0	0,63	0	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
1040	30,2	56	2432	0	0,52	0	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
1094	32	57	3874	0	0,7	0,85	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1130	31.3	78.4	434	0	0,63	0	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
1253	30,2	96,9	2432	0,06	0,52	0	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
1285	32	99,9	3874	0,11	0,7	0,85	Penyiram <i>OFF</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
2101	18	33	434	0,87	0	0.85	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1345	19	35	2432	0,18	0	0.75	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1872	17	31	3874	0,84	0	0,95	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
2101	15	55,4	434	0,87	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>

Parameter				Nilai Fuzzy			Aktuator
Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Penyiram	Kipas	Lampu	
2101	18	56	2432	0,87	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1345	19	57	3874	0,18	0	0,85	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1872	17	92.7	767	0,84	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
2101	15	81,1	2736	0,87	0	0,09	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
2101	18	80,6	3905	0.87	0	0.87	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1345	28,2	33	767	0.18	0,319	0.85	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1872	28,2	36	2736	0.84	0.32	0.7	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
2101	28,1	30	3905	0.87	0.31	1	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
1345	28,2	55,4	767	0.18	0.32	0	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1872	28,2	56	2736	0.84	0.32	0,09	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
2101	28,1	57	3905	0.87	0.31	0.87	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
2101	28,1	92.7	767	0,87	0,31	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1345	27,7	81,1	2736	0,18	0,27	0,09	Penyiram <i>OFF</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
1872	28,1	80,6	3905	0,84	0,31	0,81	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
2379	30,6	33	882	0.53	0,56	0.85	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1721	31,3	35	1651	0.65	0.63	0.75	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>

Parameter				Nilai Fuzzy			Aktuator
Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Penyiram	Kipas	Lampu	
2354	30,5	31	3874	0.56	0,55	0,95	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1914	31,1	55,4	882	0.89	0.61	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
2379	30,6	56	1651	0.53	0.56	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
2379	30,6	73,4	3760	0,56	0,56	0,77	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
1721	31,3	99,9	882	0,65	0,63	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
2354	30,5	79,1	1651	0,56	0,55	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
1914	31,1	99,9	3874	0.89	0,63	0,85	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
3464	15	33	882	0,58	0	0.85	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3469	18	35	1651	0,58	0	0.75	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3549	19	31	3874	0,63	0	0,95	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3464	17	55,4	882	0,58	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
3469	15	56	1651	0.58	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
3549	15	55,4	3874	0,63	0	0,85	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
4037	18	74.8	882	0,96	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
4037	19	74.9	1651	0,96	0	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>

Parameter				Nilai Fuzzy			Aktuator
Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Penyiram	Kipas	Lampu	
3464	17	75.2	3874	0,58	0	0,85	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3469	28.3	33	384	0,58	0,33	0.85	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3549	28.3	35	2193	0,63	0,33	0.75	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
4037	28.3	31	3517	0,96	0,33	0,95	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
4037	28.3	55,4	384	0,96	0,33	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
3464	28.3	56	2193	0,58	0,33	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
3469	28.3	55,4	3517	0,58	0,33	0,61	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3464	28.3	74.8	384	0.58	0.33	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
3469	28.3	74.9	2193	0.58	0,33	0	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>OFF</i>
3549	28.3	75.2	3517	0.63	0,33	0,61	Penyiram <i>ON</i> , Kipas <i>OFF</i> , Lampu <i>ON</i>
3549	30,6	33	384	0,63	0,56	0.85	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
4037	30,7	35	2193	0,96	0,57	0.75	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
4037	30,6	31	3517	0,96	0,56	0,95	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
3464	31,3	55,4	384	0,58	0,63	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
3469	30,5	56	1664	0,58	0,55	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>

Parameter				Nilai Fuzzy			Aktuator
Kelembapan Tanah	Suhu Udara	Kelembapan Udara	Intensitas Cahaya	Penyiram	Kipas	Lampu	
3549	31,1	55,4	3517	0,63	0,61	0,61	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>
4037	31,3	99	384	0,96	0,63	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
4037	30,6	99,9	1664	0,96	0,56	0	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>OFF</i>
4037	30,7	99,9	3517	0,96	0,57	0,61	Penyiram <i>ON</i> , Kipas <i>ON</i> , Lampu <i>ON</i>

Lampiran 3 : Hasil Latensi Antara Mikrokontroler dan Server pada Skenario Pertama

Waktu Kirim	Waktu Tiba	Latensi
07:14:00,736	07:14:00,838	102
07:14:15,736	07:14:15,838	102
07:14:30,736	07:14:30,837	101
07:14:45,736	07:14:45,839	103
07:15:00,736	07:15:00,837	101
07:15:15,736	07:15:15,837	101
07:15:30,736	07:15:30,837	101
07:15:45,736	07:15:45,839	103
07:16:00,736	07:16:00,837	101
07:16:15,736	07:16:15,838	102
07:16:30,736	07:16:30,867	131
07:16:45,736	07:16:45,840	104
07:17:00,736	07:17:00,839	103
07:17:15,736	07:17:15,840	104
07:17:30,736	07:17:30,840	104
07:17:45,736	07:17:45,840	104
07:18:00,736	07:18:00,849	113
07:18:15,736	07:18:15,844	108
07:18:30,736	07:18:30,839	103
07:18:45,736	07:18:45,839	103
07:19:00,736	07:19:00,835	99
07:19:15,736	07:19:15,839	103
07:19:30,736	07:19:30,836	100
07:19:45,736	07:19:45,837	101
07:20:00,736	07:20:00,839	103
07:20:15,736	07:20:15,839	103
07:20:30,736	07:20:30,838	102
07:20:45,736	07:20:45,838	102
07:21:00,736	07:21:00,836	100
07:21:15,736	07:21:15,838	102
07:21:30,736	07:21:30,836	100
07:21:45,736	07:21:45,836	100
07:22:00,736	07:22:00,837	101
07:22:15,736	07:22:15,838	102
07:22:30,736	07:22:30,836	100
07:22:45,736	07:22:45,837	101
07:23:00,736	07:23:00,841	105
07:23:15,736	07:23:15,839	103
07:23:30,736	07:23:30,837	101

Waktu Kirim	Waktu Tiba	Latensi
07:23:45,736	07:23:45,837	101
07:24:00,736	07:24:00,847	111
07:24:15,736	07:24:15,862	126
07:24:30,736	07:24:30,849	113
07:24:45,736	07:24:45,847	111
07:25:00,736	07:25:00,856	120
07:25:15,736	07:25:15,847	111
07:25:30,736	07:25:30,862	126
07:25:45,736	07:25:45,847	111
07:26:00,736	07:26:00,851	115
07:26:15,736	07:26:15,849	113
07:26:30,736	07:26:30,847	111
07:26:45,736	07:26:45,852	116
07:27:00,736	07:27:00,861	125
07:27:15,736	07:27:15,864	128
07:27:30,736	07:27:30,849	113
07:27:45,736	07:27:45,851	115
07:28:00,736	07:28:00,853	117
07:28:15,736	07:28:15,849	113
07:28:30,736	07:28:30,847	111
07:28:45,736	07:28:45,848	112

Lampiran 4: Hasil Latensi Antara Mikrokontroler dan Website pada Skenario pertama

Waktu Kirim	Waktu Tiba	Latensi
07:14:00,838	07:14:02,531	1693
07:14:15,838	07:14:22,022	6184
07:14:30,837	07:14:33,096	2259
07:14:45,839	07:14:47,404	1565
07:15:00,837	07:15:04,808	3971
07:15:15,837	07:15:17,290	1453
07:15:30,837	07:15:36,740	5903
07:15:45,839	07:15:54,768	8929
07:16:00,837	07:16:05,681	4844
07:16:15,838	07:16:18,631	2793
07:16:30,867	07:16:32,456	1589
07:16:45,840	07:16:48,215	2375
07:17:00,839	07:17:07,702	6863
07:17:15,840	07:17:17,422	1582
07:17:30,840	07:17:32,387	1547

Waktu Kirim	Waktu Tiba	Latensi
07:17:45,840	07:17:47,412	1572
07:18:00,849	07:18:02,503	1654
07:18:15,844	07:18:17,359	1515
07:18:30,839	07:18:34,704	3865
07:18:45,839	07:18:49,960	4121
07:19:00,835	07:19:02,488	1653
07:19:15,839	07:19:23,189	7350
07:19:30,836	07:19:32,570	1734
07:19:45,837	07:19:47,492	1655
07:20:00,839	07:20:06,283	5444
07:20:15,839	07:20:20,068	4229
07:20:30,838	07:20:33,660	2822
07:20:45,838	07:20:50,253	4415
07:21:00,836	07:21:02,365	1529
07:21:15,838	07:21:18,869	3031
07:21:30,836	07:21:35,033	4197
07:21:45,836	07:21:52,430	6594
07:22:00,837	07:22:06,079	5242
07:22:15,838	07:22:17,393	1555
07:22:30,836	07:22:32,336	1500
07:22:45,837	07:22:47,458	1621
07:23:00,841	07:23:02,379	1538
07:23:15,839	07:23:17,398	1559
07:23:30,837	07:23:32,375	1538
07:23:45,837	07:23:47,336	1499
07:24:00,847	07:24:08,789	7942
07:24:15,862	07:24:17,557	1695
07:24:30,849	07:24:40,492	9643
07:24:45,847	07:24:47,408	1561
07:25:00,856	07:25:02,659	1803
07:25:15,847	07:25:19,693	3846
07:25:30,862	07:25:32,596	1734
07:25:45,847	07:25:50,002	4155
07:26:00,851	07:26:03,011	2160
07:26:15,849	07:26:17,753	1904
07:26:30,847	07:26:32,346	1499
07:26:45,852	07:26:49,939	4087
07:27:00,861	07:27:11,465	10604
07:27:15,864	07:27:20,074	4210
07:27:30,849	07:27:32,519	1670
07:27:45,851	07:27:49,070	3219

Waktu Kirim	Waktu Tiba	Latensi
07:28:00,853	07:28:02,818	1965
07:28:15,849	07:28:17,527	1678
07:28:30,847	07:28:38,221	7374
07:28:45,848	07:28:51,282	5434

Lampiran 5: Hasil Latensi Antara Mikrokontroler dan Server pada Skenario Kedua

Waktu Kirim	Waktu Tiba	Latensi
08:42:51,987	08:42:53,650	1663
08:43:06,996	08:43:08,657	1661
08:43:22,005	08:43:26,166	4161
08:43:37,016	08:43:44,854	7838
08:43:52,025	08:43:53,636	1611
08:44:07,034	08:44:08,771	1737
08:44:22,043	08:44:23,649	1606
08:44:37,054	08:44:38,647	1593
08:44:52,063	08:44:54,445	2382
08:45:07,072	08:45:09,310	2238
08:45:22,081	08:45:26,035	3954
08:45:37,092	08:45:38,793	1701
08:45:52,101	08:45:53,788	1687
08:46:07,110	08:46:08,886	1776
08:46:22,119	08:46:23,750	1631
08:46:37,130	08:46:39,006	1876
08:46:52,148	08:46:54,627	2479
08:47:07,157	08:47:09,698	2541
08:47:22,166	08:47:23,862	1696
08:47:37,175	08:47:38,810	1635
08:47:52,185	08:47:54,170	1985
08:48:07,194	08:48:08,813	1619
08:48:22,203	08:48:23,868	1665
08:48:37,212	08:48:40,016	2804
08:48:52,223	08:48:54,230	2007
08:49:07,233	08:49:08,928	1695
08:49:22,242	08:49:23,892	1650
08:49:37,251	08:49:38,893	1642
08:49:52,262	08:49:53,994	1732
08:50:07,271	08:50:09,001	1730
08:50:22,280	08:50:23,932	1652
08:50:37,289	08:50:38,979	1690

Waktu Kirim	Waktu Tiba	Latensi
08:50:52,300	08:50:54,047	1747
08:51:07,309	08:51:09,013	1704
08:51:22,318	08:51:23,790	1472
08:51:37,327	08:51:39,073	1746
08:51:52,338	08:51:54,927	2589
08:52:07,347	08:52:09,050	1703
08:52:22,356	08:52:23,905	1549
08:52:37,365	08:52:38,874	1509
08:52:52,376	08:52:53,910	1534
08:53:07,385	08:53:08,954	1569
08:53:22,394	08:53:23,982	1588
08:53:37,403	08:53:38,968	1565
08:53:52,415	08:53:53,928	1513
08:54:07,424	08:54:08,966	1542
08:54:22,433	08:54:23,954	1521
08:54:37,442	08:54:39,498	2056
08:54:52,453	08:54:53,968	1515
08:55:07,462	08:55:09,007	1545
08:55:22,471	08:55:23,982	1511
08:55:37,481	08:55:39,052	1571
08:55:52,492	08:55:54,062	1570
08:56:07,501	08:56:09,006	1505
08:56:22,510	08:56:24,085	1575
08:56:37,519	08:56:39,128	1609
08:56:52,530	08:56:54,076	1546
08:57:07,539	08:57:09,089	1550
08:57:22,548	08:57:25,054	2506
08:57:37,557	08:57:39,096	1539

Lampiran 6: Hasil Latensi Antara Mikrokontroler dan Website pada Skenario Kedua

Waktu Kirim	Waktu Tiba	Latensi
08:42:51,987	08:42:54,750	2763
08:43:06,996	08:43:09,220	2224
08:43:22,005	08:43:28,079	6074
08:43:37,016	08:43:45,902	8886
08:43:52,025	08:43:54,282	2257
08:44:07,034	08:44:10,022	2988
08:44:22,043	08:44:24,575	2532
08:44:37,054	08:44:40,123	3069

08:44:52,063	08:44:55,054	2991
08:45:07,072	08:45:10,062	2990
08:45:22,081	08:45:27,852	5771
08:45:37,092	08:45:39,318	2226
08:45:52,101	08:45:54,341	2240
08:46:07,110	08:46:10,503	3393
08:46:22,119	08:46:24,270	2151
08:46:37,130	08:46:40,545	3415
08:46:52,148	08:46:55,887	3739
08:47:07,157	08:47:11,004	3847
08:47:22,166	08:47:24,445	2279
08:47:37,175	08:47:40,314	3139
08:47:52,185	08:47:54,667	2482
08:48:07,194	08:48:09,352	2158
08:48:22,203	08:48:25,490	3287
08:48:37,212	08:48:41,424	4212
08:48:52,223	08:48:54,803	2580
08:49:07,233	08:49:09,521	2288
08:49:22,242	08:49:25,259	3017
08:49:37,251	08:49:39,483	2232
08:49:52,262	08:49:55,369	3107
08:50:07,271	08:50:10,962	3691
08:50:22,280	08:50:24,659	2379
08:50:37,289	08:50:40,566	3277
08:50:52,300	08:50:54,640	2340
08:51:07,309	08:51:10,235	2926
08:51:22,318	08:51:24,487	2169
08:51:37,327	08:51:39,908	2581
08:51:52,338	08:51:55,608	3270
08:52:07,347	08:52:10,903	3556
08:52:22,356	08:52:24,609	2253
08:52:37,365	08:52:39,543	2178
08:52:52,376	08:52:54,619	2243
08:53:07,385	08:53:09,640	2255
08:53:22,394	08:53:24,656	2262
08:53:37,403	08:53:39,584	2181
08:53:52,415	08:53:54,707	2292
08:54:07,424	08:54:09,626	2202
08:54:22,433	08:54:24,542	2109
08:54:37,442	08:54:41,264	3822
08:54:52,453	08:54:54,692	2239
08:55:07,462	08:55:11,060	3598
08:55:22,471	08:55:24,588	2117

08:55:37,481	08:55:39,713	2232
08:55:52,492	08:55:54,727	2235
08:56:07,501	08:56:09,667	2166
08:56:22,510	08:56:24,752	2242
08:56:37,519	08:56:40,893	3374
08:56:52,530	08:56:54,724	2194
08:57:07,539	08:57:11,063	3524
08:57:22,548	08:57:25,756	3208
08:57:37,557	08:57:39,759	2202

Lampiran 7: Hasil Latensi Antara Mikrokontroler dan Server pada Skenario Ketiga

Waktu Kirim	Waktu Tiba	Latensi
08:08:47,045	08:08:47,152	107
08:09:02,045	08:09:02,152	107
08:09:17,045	08:09:17,152	107
08:09:32,045	08:09:32,155	110
08:09:47,045	08:09:47,154	109
08:10:02,045	08:10:02,153	108
08:10:17,045	08:10:17,154	109
08:10:32,045	08:10:32,154	109
08:10:47,045	08:10:47,153	108
08:11:02,045	08:11:02,154	109
08:11:17,045	08:11:17,153	108
08:11:32,045	08:11:32,151	106
08:11:47,045	08:11:47,152	107
08:12:02,045	08:12:02,153	108
08:12:17,045	08:12:17,152	107
08:12:32,045	08:12:32,151	106
08:12:47,045	08:12:47,151	106
08:13:02,045	08:13:02,155	110
08:13:17,045	08:13:17,152	107
08:13:32,045	08:13:32,155	110
08:13:47,045	08:13:47,152	107
08:14:02,045	08:14:02,152	107
08:14:17,045	08:14:17,154	109
08:14:32,045	08:14:32,154	109
08:14:47,045	08:14:47,152	107
08:15:02,045	08:15:02,152	107
08:15:17,045	08:15:17,152	107
08:15:32,045	08:15:32,152	107
08:15:47,045	08:15:47,153	108

Waktu Kirim	Waktu Tiba	Latensi
08:16:02,045	08:16:02,157	112
08:16:17,045	08:16:17,120	75
08:16:32,045	08:16:32,122	77
08:16:47,045	08:16:47,123	78
08:17:02,045	08:17:02,121	76
08:17:17,045	08:17:17,121	76
08:17:32,045	08:17:32,123	78
08:17:47,045	08:17:47,124	79
08:18:02,045	08:18:02,122	77
08:18:17,045	08:18:17,122	77
08:18:32,045	08:18:32,121	76
08:18:47,045	08:18:47,121	76
08:19:02,045	08:19:02,121	76
08:19:17,045	08:19:17,123	78
08:19:32,045	08:19:32,123	78
08:19:47,045	08:19:47,125	80
08:20:02,045	08:20:02,125	80
08:20:17,045	08:20:17,123	78
08:20:32,045	08:20:32,122	77
08:20:47,045	08:20:47,123	78
08:21:02,045	08:21:02,122	77
08:21:17,045	08:21:17,157	112
08:21:32,045	08:21:32,156	111
08:21:47,045	08:21:47,156	111
08:22:02,045	08:22:02,156	111
08:22:17,045	08:22:17,159	114
08:22:32,045	08:22:32,162	117
08:22:47,045	08:22:47,156	111
08:23:02,045	08:23:02,160	115
08:23:17,045	08:23:17,158	113
08:23:32,045	08:23:32,156	111

Lampiran 8: Hasil Latensi Antara Mikrokontroler dan Website pada Skenario Ketiga

Waktu Kirim	Waktu Tiba	Latensi
08:08:47,045	08:09:00,174	1040
08:09:02,045	08:09:03,234	1037
08:09:17,045	08:09:18,255	986
08:09:32,045	08:09:33,279	1030
08:09:47,045	08:09:48,259	1013

Waktu Kirim	Waktu Tiba	Latensi
08:09:47,045	08:10:03,275	1023
08:10:17,045	08:10:18,270	1097
08:10:32,045	08:10:33,246	1034
08:10:32,045	08:10:48,246	1033
08:11:02,045	08:11:03,223	982
08:11:17,045	08:11:18,257	980
08:11:32,045	08:11:33,220	942
08:11:47,045	08:11:48,219	1007
08:12:02,045	08:12:03,255	1099
08:12:17,045	08:12:18,255	1204
08:12:32,045	08:12:33,236	1024
08:12:32,045	08:12:48,281	1058
08:13:02,045	08:13:03,266	1040
08:13:17,045	08:13:18,259	1082
08:13:32,045	08:13:33,238	988
08:13:47,045	08:13:48,228	1102
08:14:02,045	08:14:03,230	1111
08:14:17,045	08:14:18,255	1009
08:14:32,045	08:14:33,232	1046
08:14:47,045	08:14:48,233	981
08:15:02,045	08:15:03,225	1044
08:15:17,045	08:15:18,260	1042
08:15:32,045	08:15:33,251	1010
08:15:47,045	08:15:48,262	1006
08:16:02,045	08:16:03,238	1027
08:16:17,045	08:16:18,242	1063
08:16:32,045	08:16:33,275	998
08:16:47,045	08:16:48,230	1025
08:17:02,045	08:17:03,258	1180
08:17:17,045	08:17:18,245	1013
08:17:32,045	08:17:33,308	1076
08:17:47,045	08:17:48,306	1098
08:18:02,045	08:18:03,261	1021
08:18:02,045	08:18:19,996	1475
08:18:32,045	08:18:33,243	1048
08:18:47,045	08:18:48,223	1005
08:19:02,045	08:19:03,245	998
08:19:17,045	08:19:18,214	998
08:19:32,045	08:19:33,224	1078
08:19:47,045	08:19:48,234	1020
08:19:47,045	08:20:03,212	1014

Waktu Kirim	Waktu Tiba	Latensi
08:20:17,045	08:20:18,202	998
08:20:32,045	08:20:33,207	939
08:20:47,045	08:20:48,215	958
08:21:02,045	08:21:03,223	1037
08:21:17,045	08:21:18,217	1010
08:21:17,045	08:21:33,212	1019
08:21:47,045	08:21:48,180	985
08:22:02,045	08:22:03,220	1103
08:22:17,045	08:22:18,195	1061
08:22:32,045	08:22:33,203	1007
08:22:47,045	08:22:48,237	1046
08:23:02,045	08:23:03,180	974
08:23:17,045	08:23:18,211	970
08:23:32,045	08:23:33,239	958

Lampiran 9: Tampilan Website untuk Pengendalian dan Pemantauan

Smart Control Dashboard



Lampiran 10: ERD Database

Kontrol		Sensor	
PK	<u>id</u>	PK	<u>id</u>
	otomatis		temperature
	kipas		humidity
	lampu		light
	penyiram		soil
	update_at		time

